



It All Starts with a Single Request.

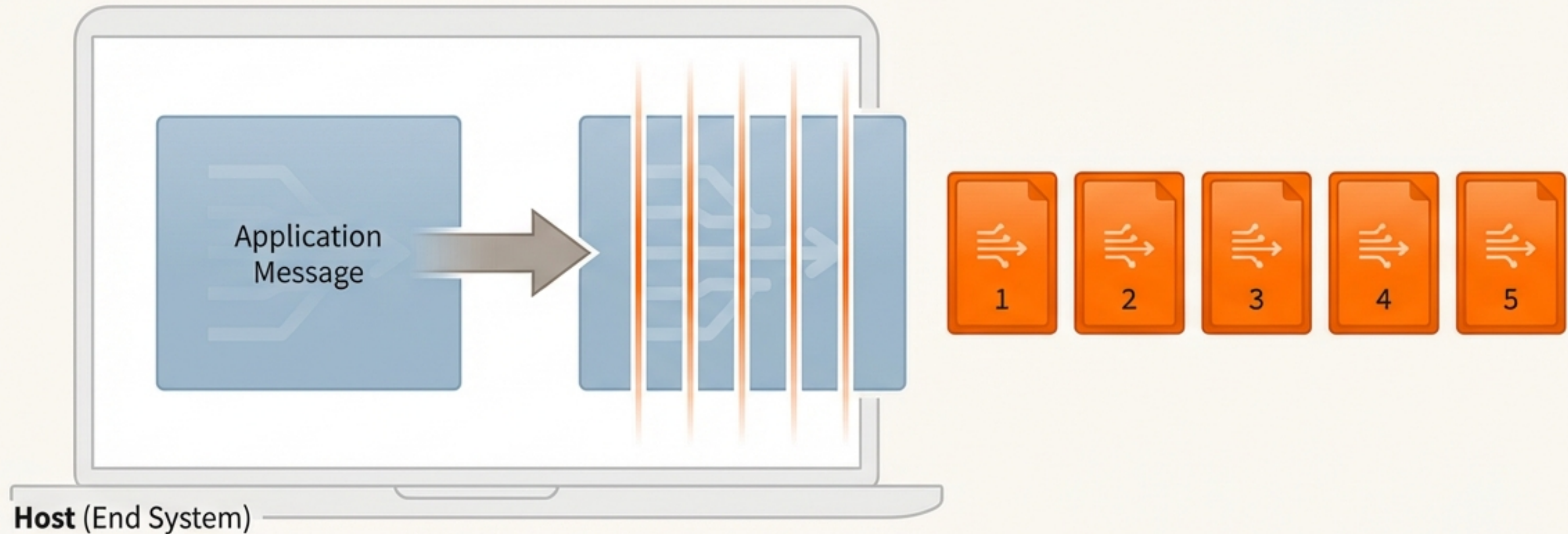
Every time you browse the web, send an email, or stream a video, you set in motion an incredible journey.

This isn't the story of the Internet as a vague cloud. This is the story of a single packet of data, from its creation to its destination.

We'll follow its path to understand the architecture of one of humanity's most complex creations.

The Internet is an infrastructure that provides services to applications we use daily (Web, streaming, email). Our story begins at the very edge of this network.

A Message is Broken into Packets

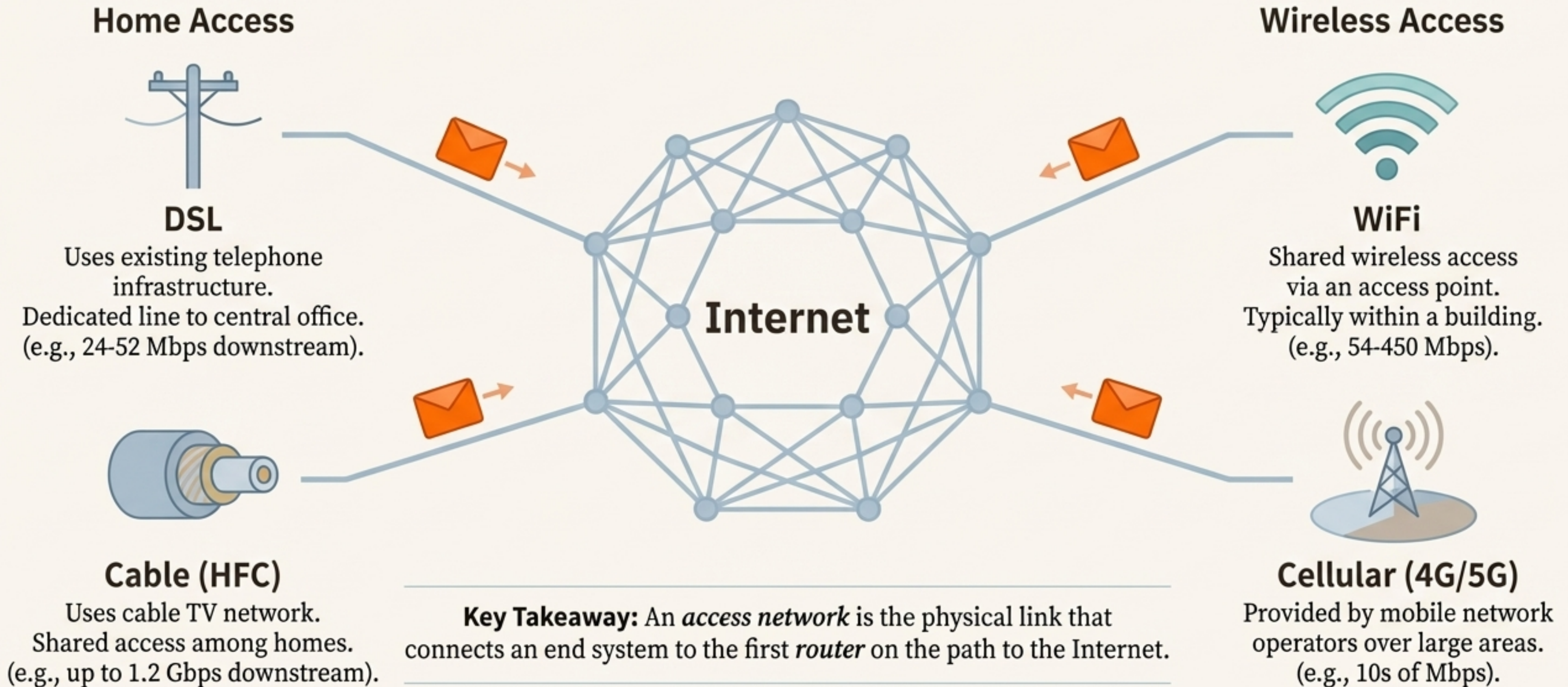


Your device, a **host** on the Internet's edge, doesn't send your request all at once. It breaks the application's message into smaller chunks known as **packets**. These packets are the fundamental units of data that travel across the Internet. Our story follows just one.

Key Terminology

Hosts (or End Systems) are the computing devices at the Internet's edge (clients and servers). *Packets* are the chunks of data they send and receive.

The On-Ramp: Connecting to the Internet.

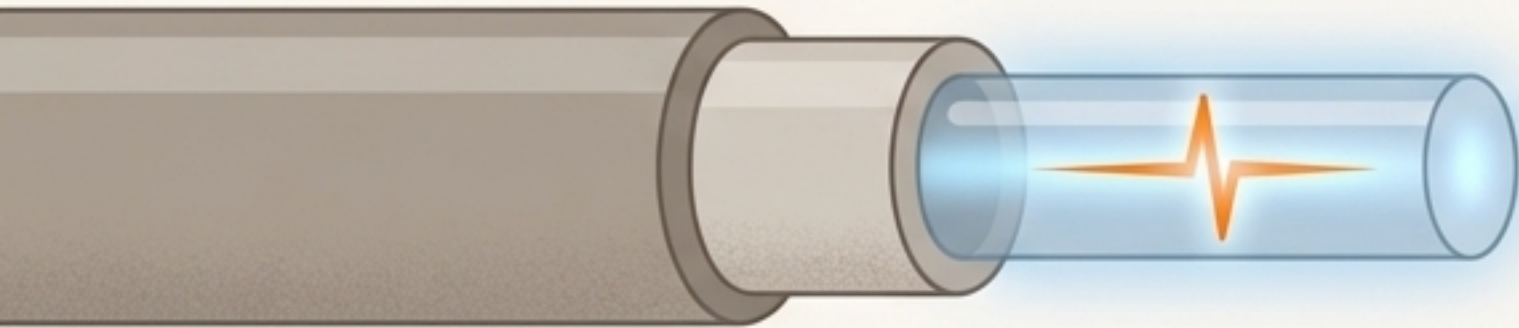


The Physical Roads of the Digital World.

Guided Media (Physical Wires)

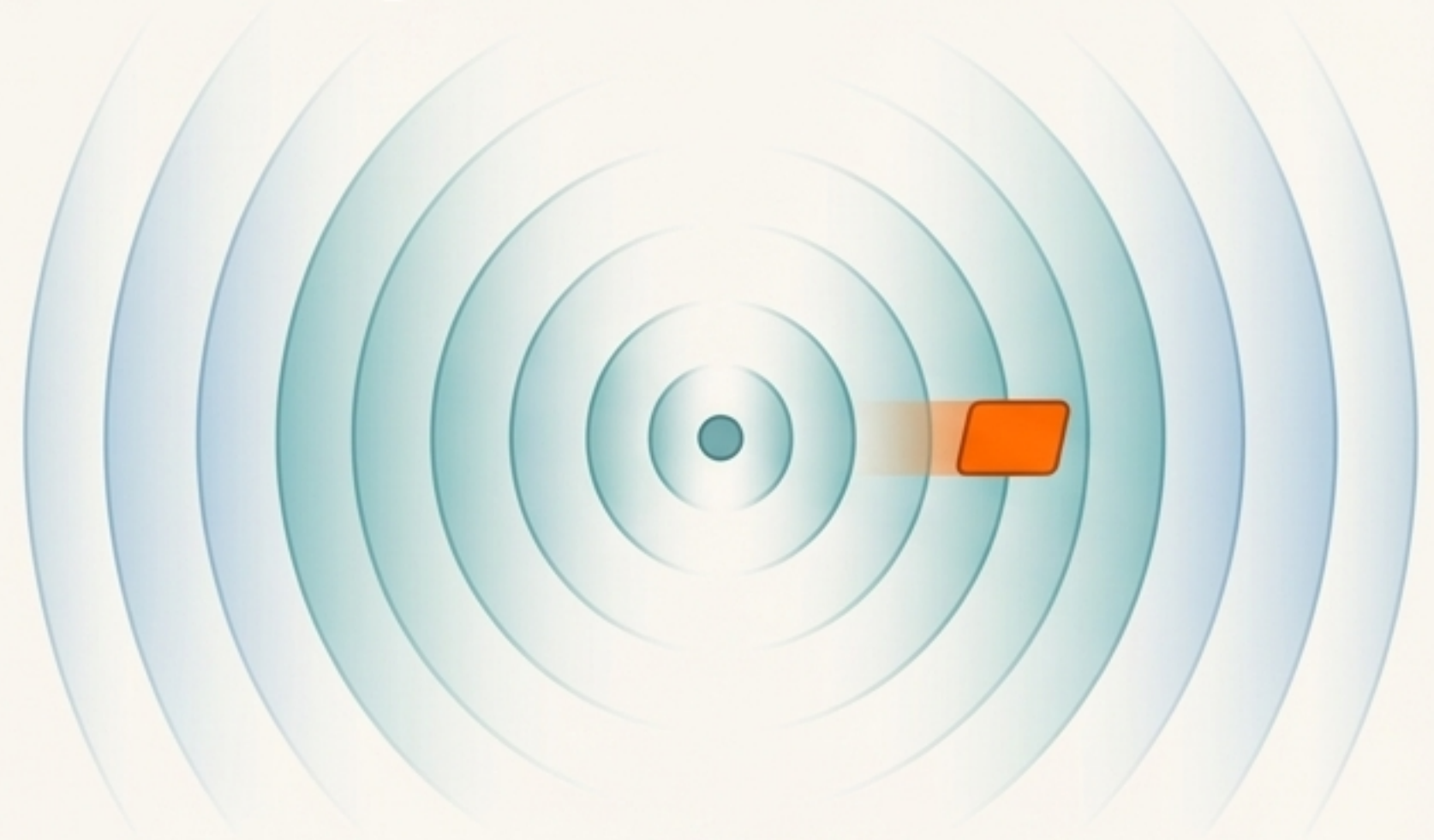


Signals propagate in solid media.
Used in Ethernet (1-10 Gbps).



Glass fiber carrying light pulses. High speed
(10s–100s Gbps) and low error rate.

Unguided Media (Wireless)



Signals propagate freely through the air (e.g., WiFi, 5G,
Satellite). Subject to interference and obstruction.

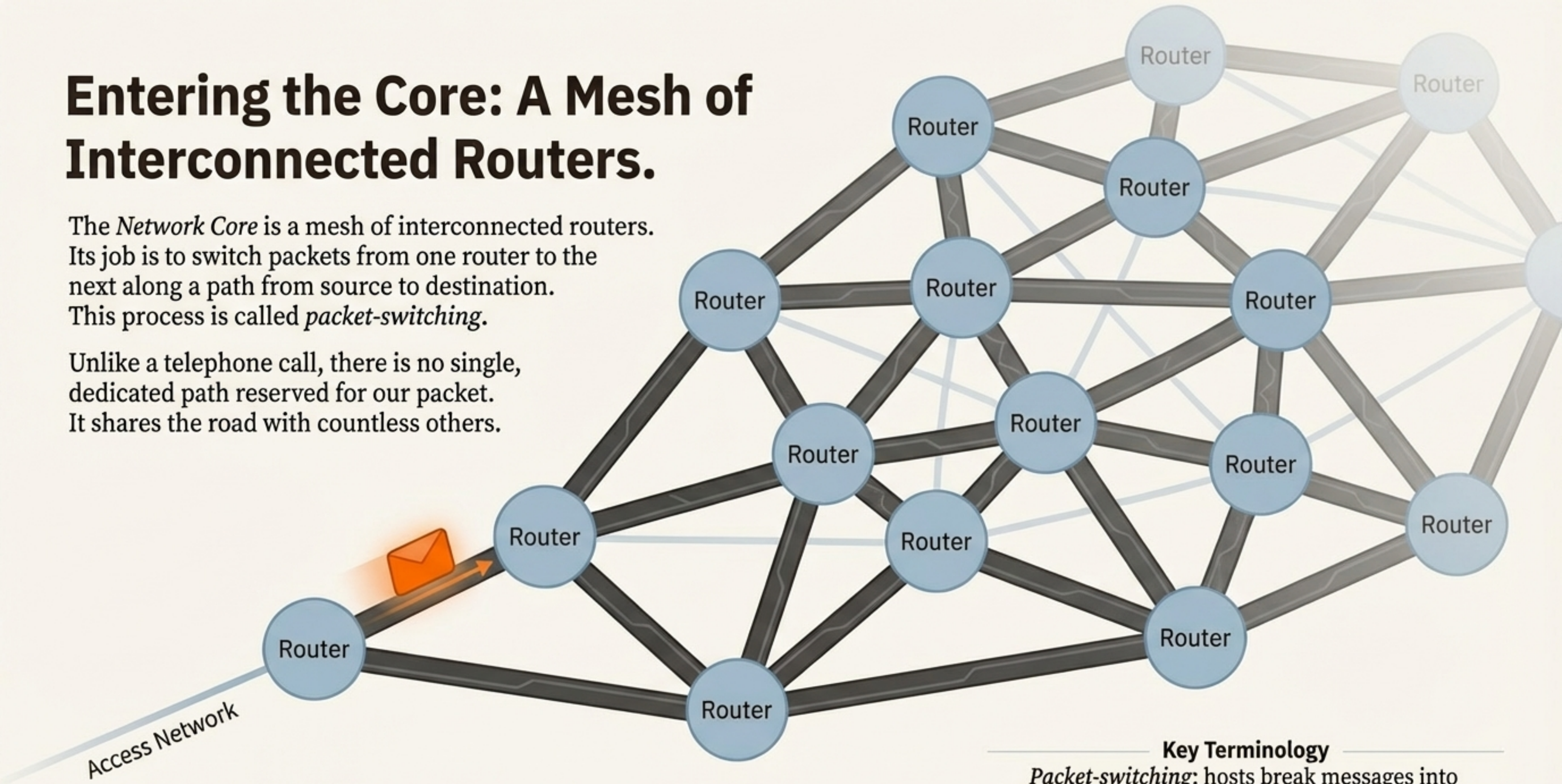
Key Takeaway

The journey relies on a diverse set of **physical media**, each with different properties and performance characteristics. The rate at which data can be sent is the **transmission rate**, or **bandwidth**.

Entering the Core: A Mesh of Interconnected Routers.

The *Network Core* is a mesh of interconnected routers. Its job is to switch packets from one router to the next along a path from source to destination. This process is called *packet-switching*.

Unlike a telephone call, there is no single, dedicated path reserved for our packet. It shares the road with countless others.

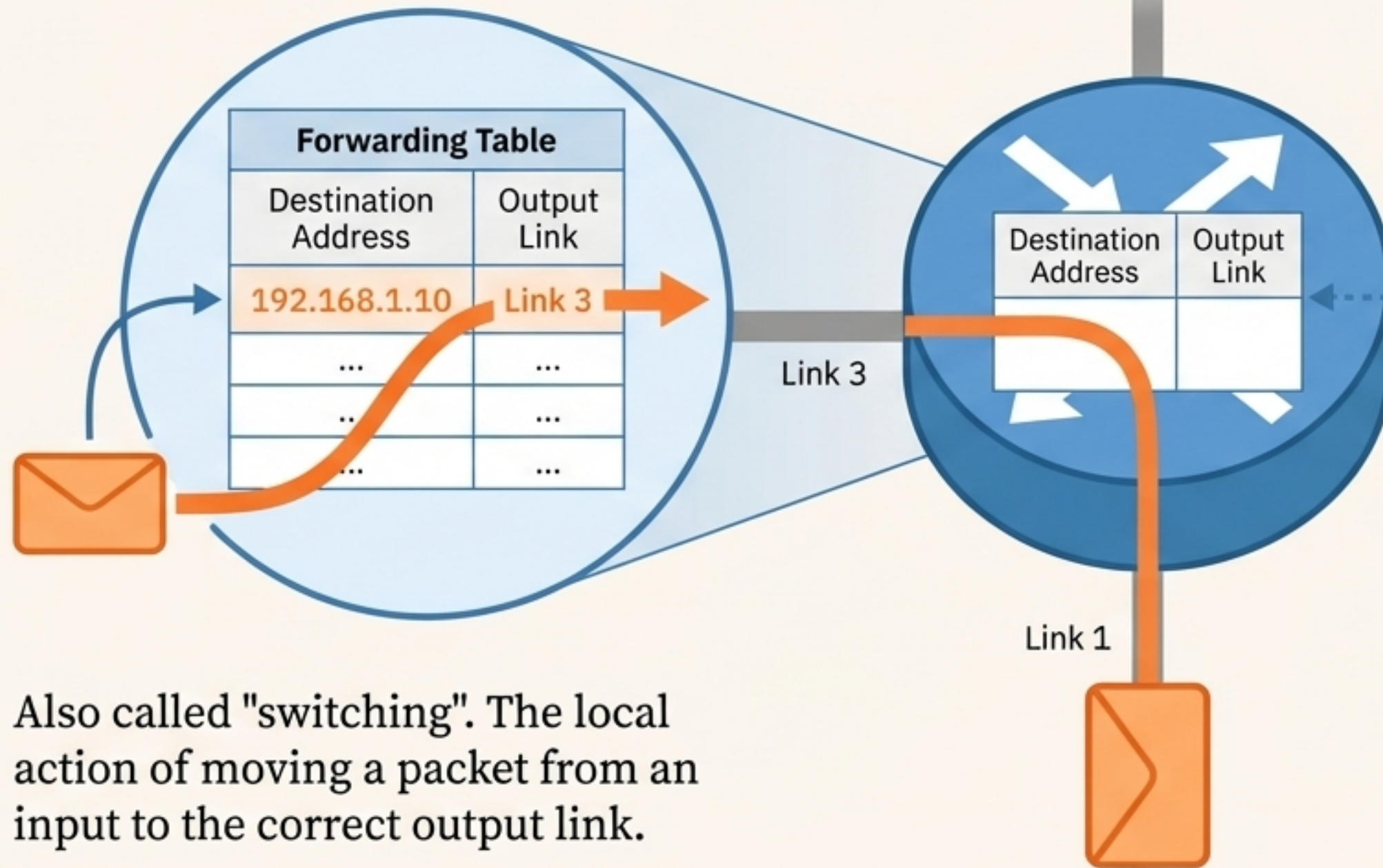


Key Terminology

Packet-switching: hosts break messages into packets, which are forwarded from router to router.

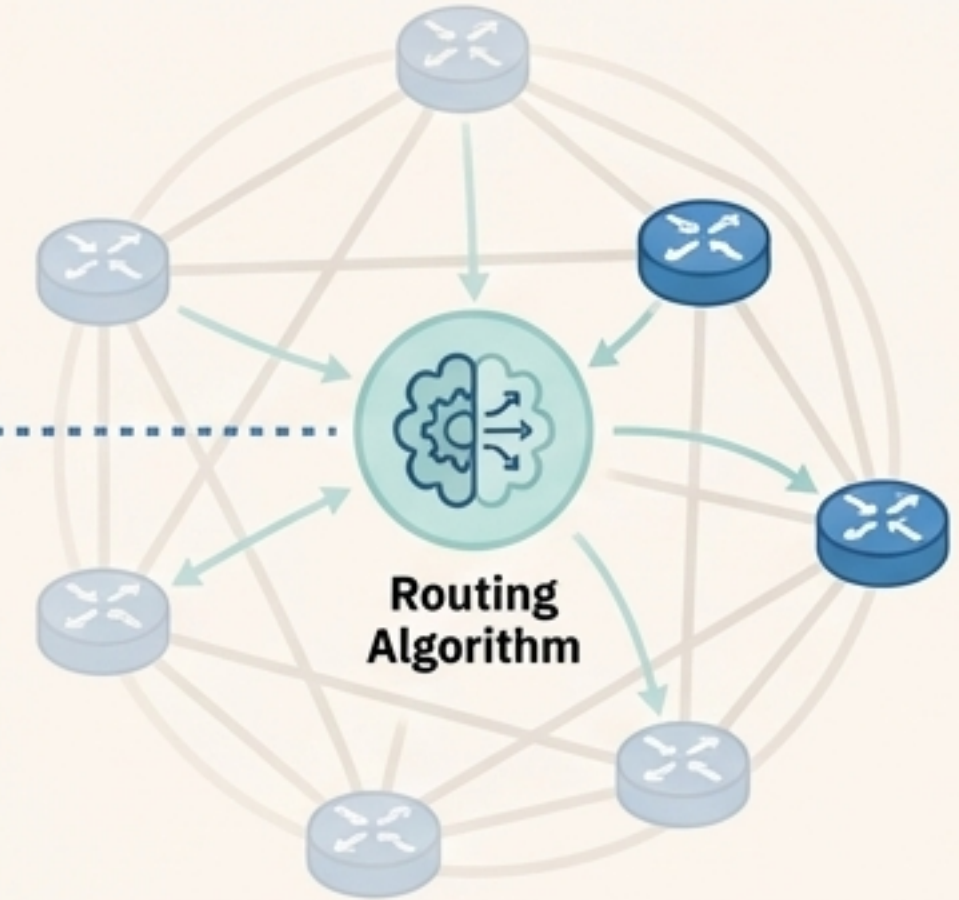
The Two Decisions at Every Crossroads

1. Forwarding (The Local Action)



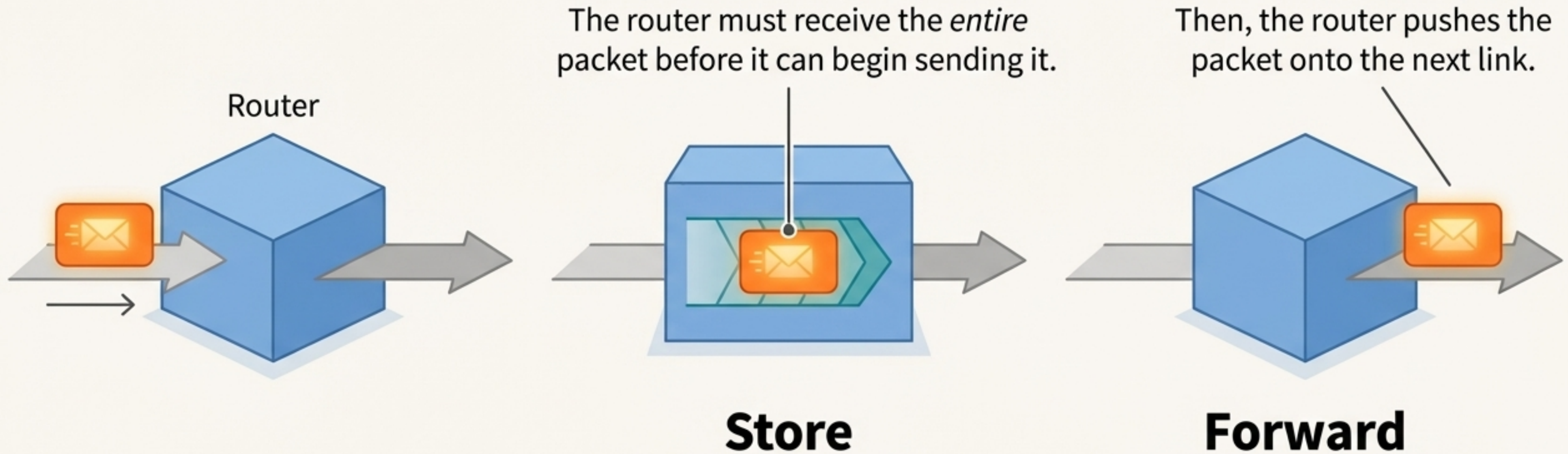
Also called "switching". The local action of moving a packet from an input to the correct output link.

2. Routing (The Global Action)



The global process of determining the paths packets take. Routing algorithms create the forwarding tables.

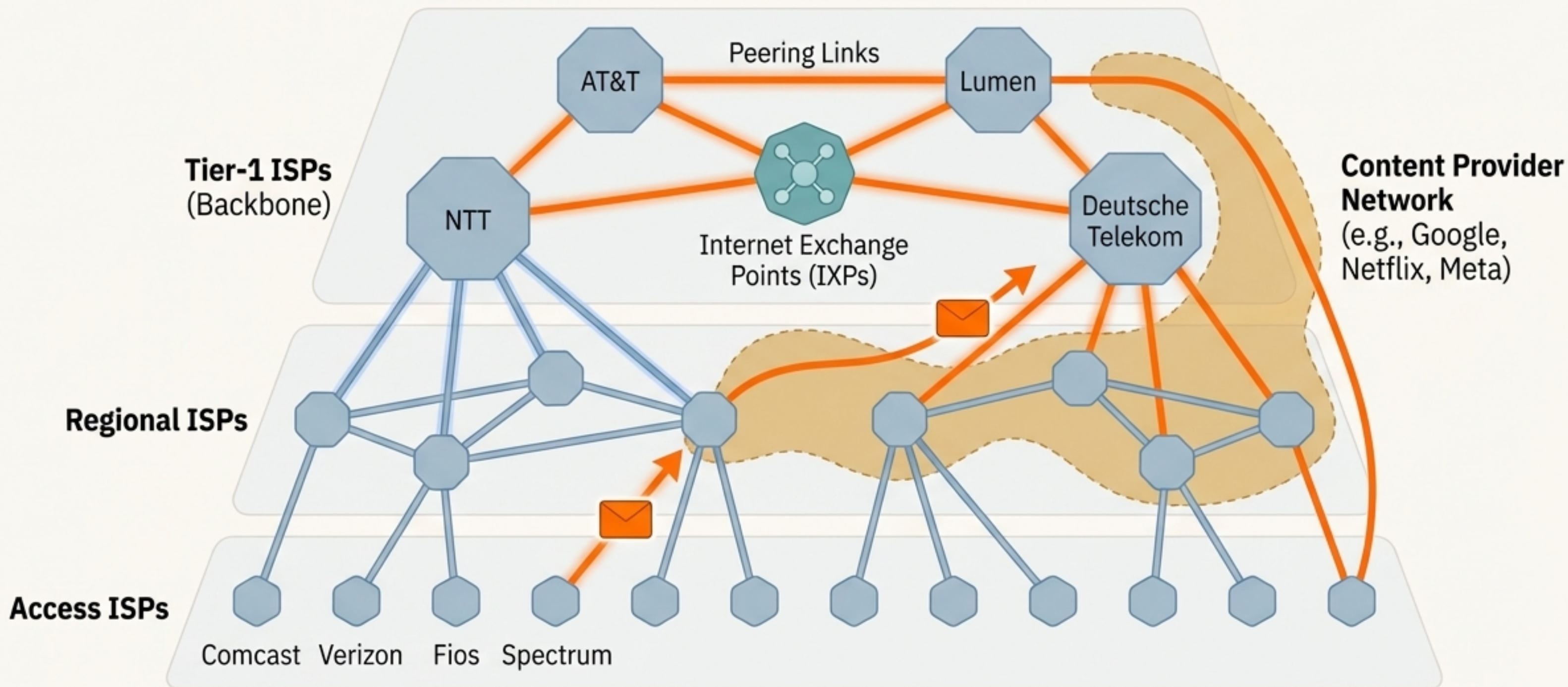
A Stop-and-Go Journey: Store-and-Forward



Packet transmission is not instantaneous. It takes time to push a packet's bits onto a link. This is the *transmission delay*: $\text{Packet Length } (L \text{ bits}) / \text{Link Rate } (R \text{ bps})$.

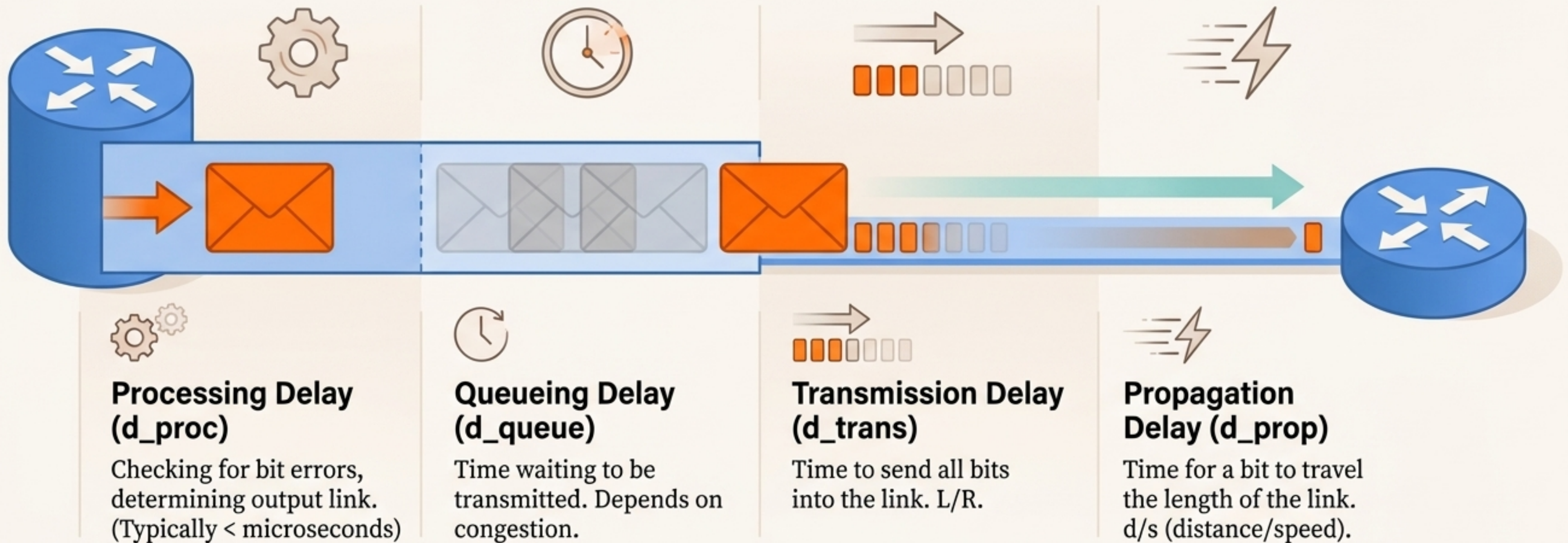
Example: For a 10 Kbit packet on a 100 Mbps link, the transmission delay is 0.1 msec.

Navigating a Network of Networks

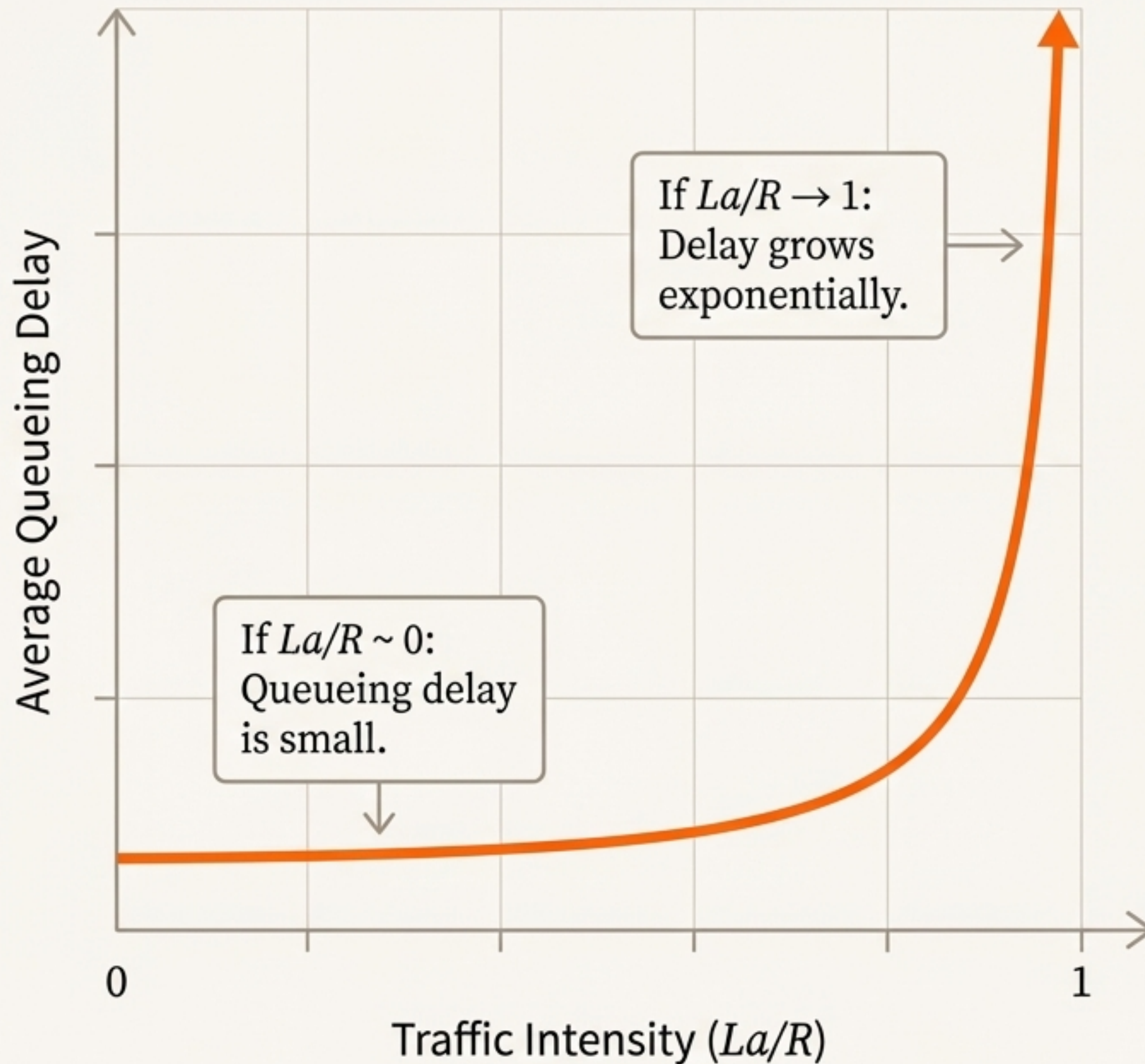


The Internet's structure is a complex, decentralized hierarchy driven by economics and policy, designed to connect any two hosts, anywhere in the world.

Why the Wait? The Four Sources of Packet Delay



$$d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$$

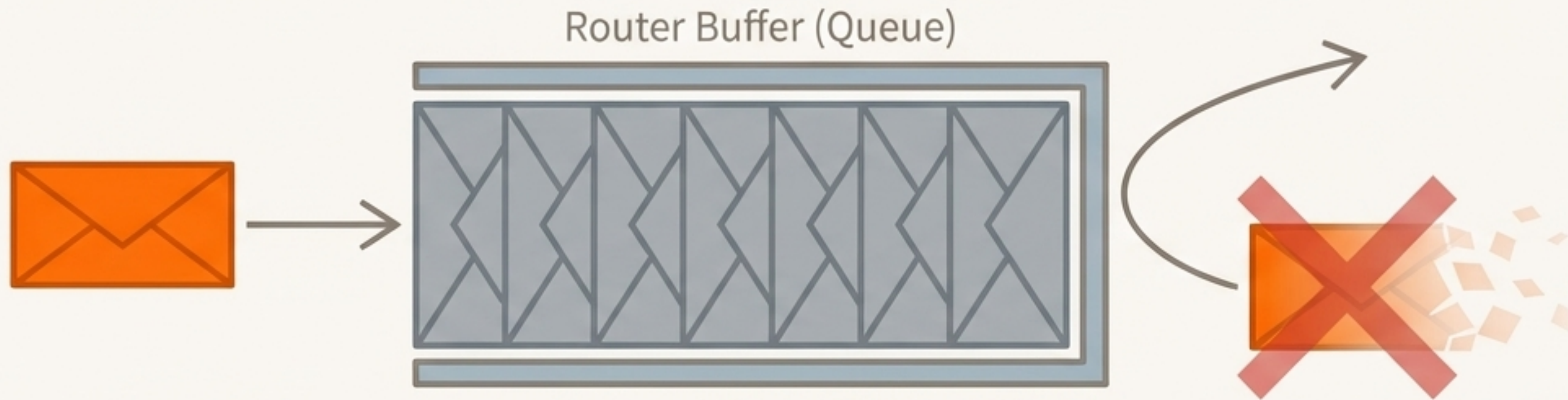


Rush Hour on the Information Superhighway

Queueing happens when packets arrive at a router faster than they can be sent out. The key metric is *Traffic Intensity*, defined as La/R , where L =packet length, a =avg. arrival rate, and R =link transmission rate.

If $La/R > 1$, more data is arriving than can be serviced; the queue will grow without bound, leading to packet loss.

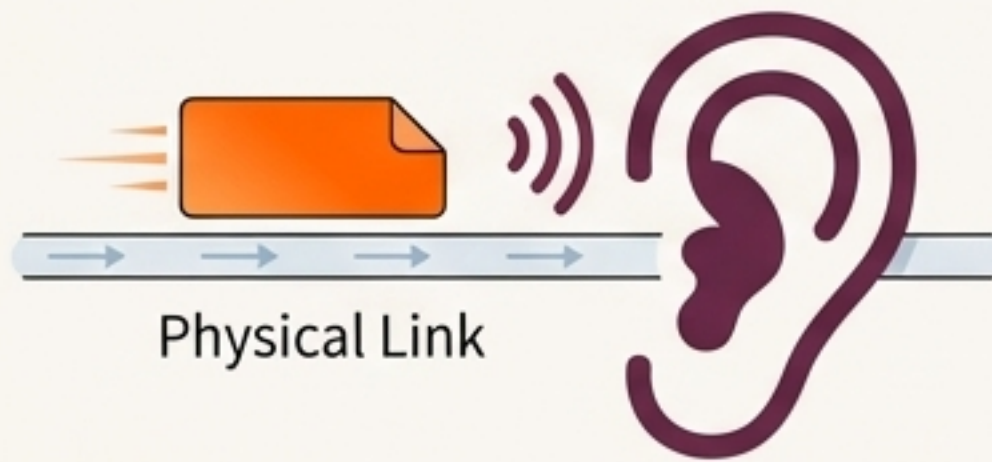
Dropped: When the Queue is Full.



Routers have finite memory, or *buffers*, to hold queued packets. When a packet arrives to find a full buffer, it has nowhere to go. The router simply *drops* the packet. This is *packet loss*. A lost packet might be retransmitted by the original sender, but this takes time and consumes more resources.

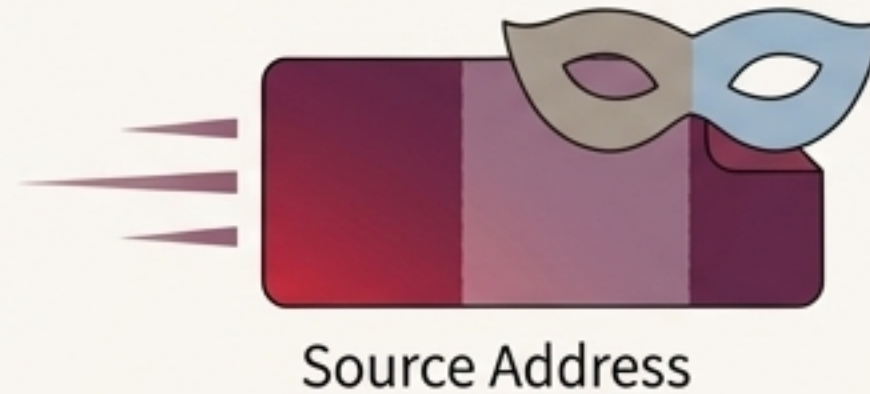
Dangers on the Digital Road

Packet Sniffing



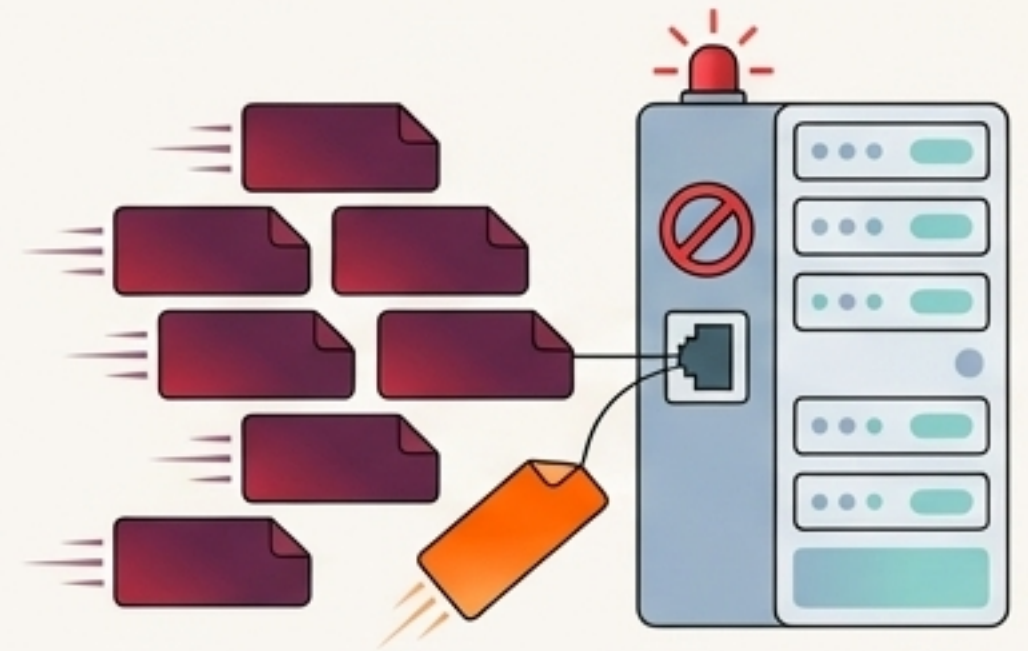
Bad guys can intercept and read the contents of packets on broadcast media (like shared WiFi)

IP Spoofing



Injecting packets with a false source address to impersonate another device.

Denial of Service (DoS)



Overwhelming a resource with bogus traffic to make it unavailable.

The Rules That Bring Order to Chaos.



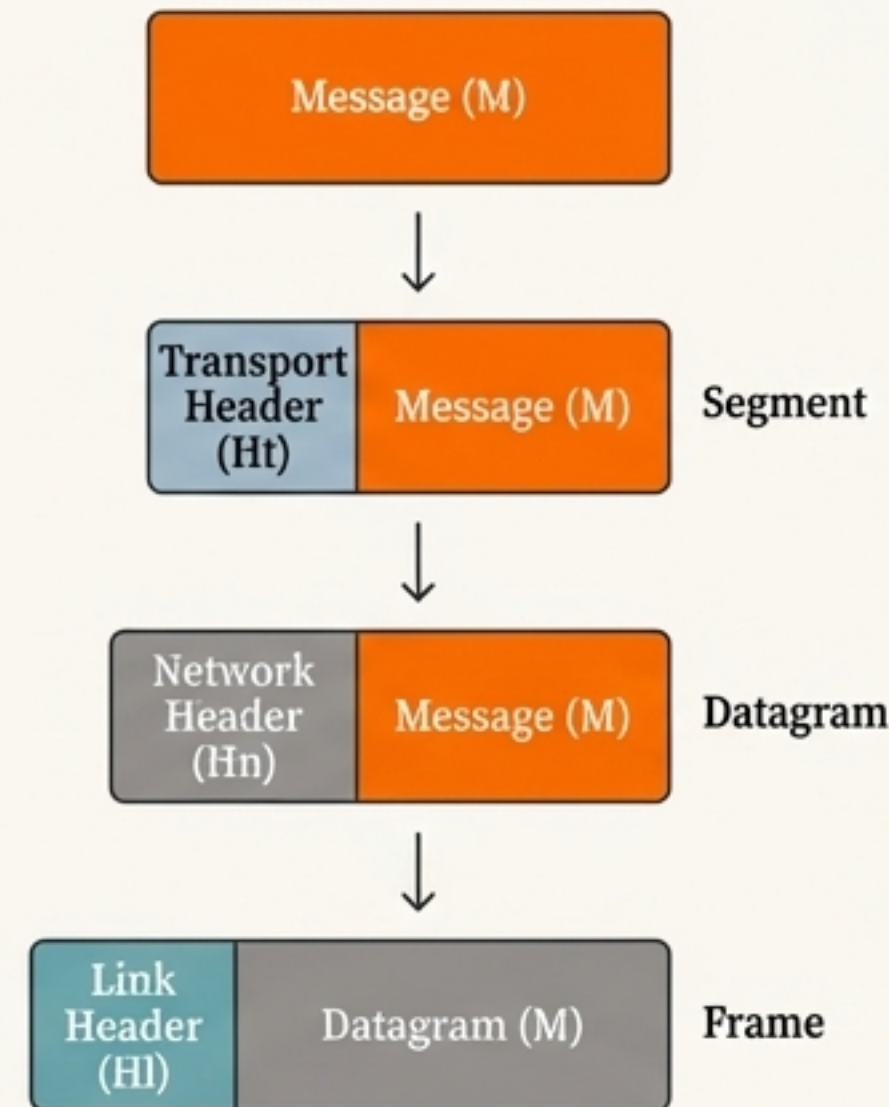
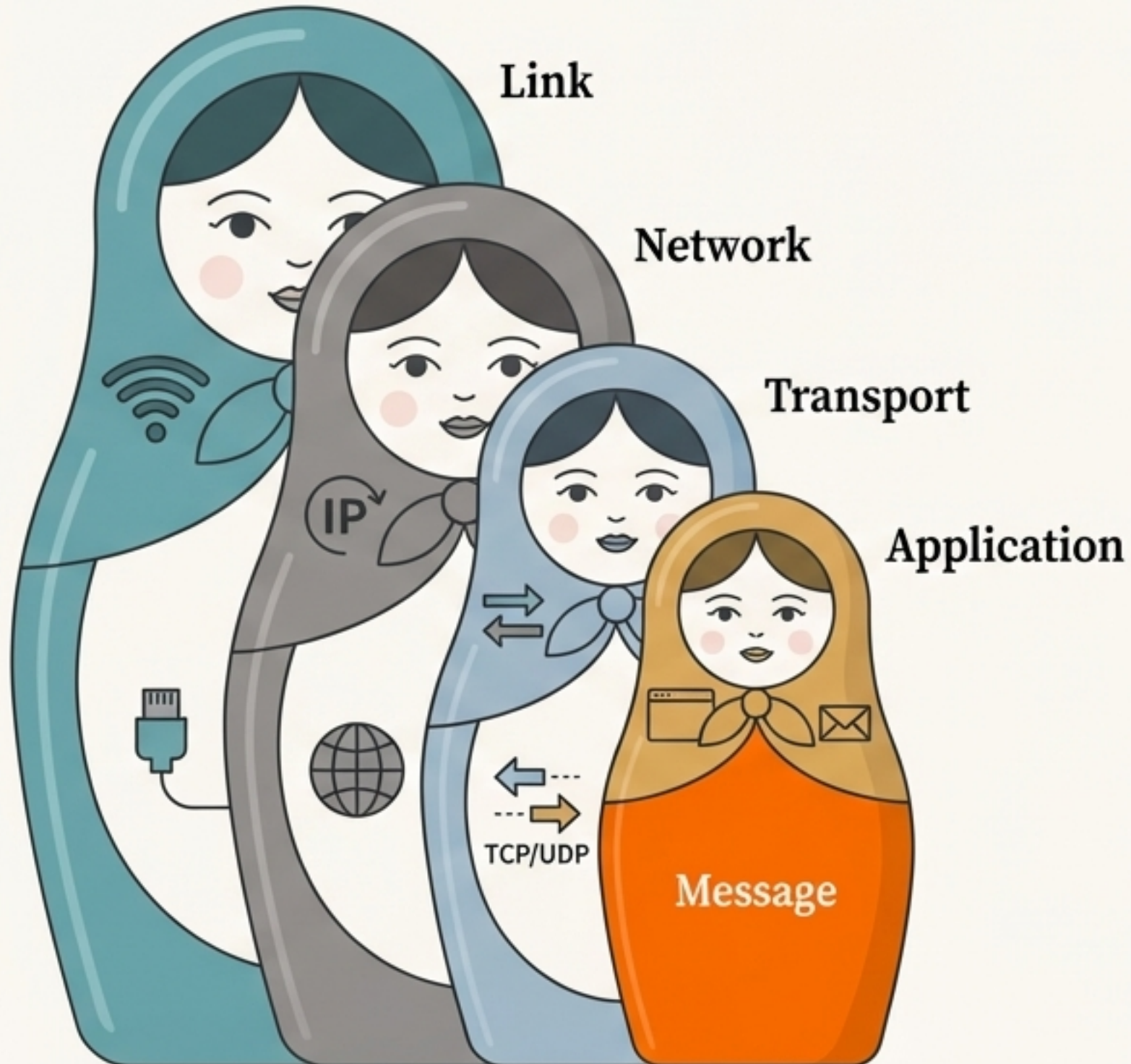
How can a complex system like the Internet function reliably?

By organizing its tasks into a series of layers. Each *layer* provides a specific service, relying on the services of the layer below it.

This is the core idea of a *protocol stack*. Protocols are the rules that govern communication at each layer, from purchasing a ticket (application) to routing the plane (network).

Layering provides modularity and a structured approach to managing network complexity.

Dressed for the Journey: Encapsulation.

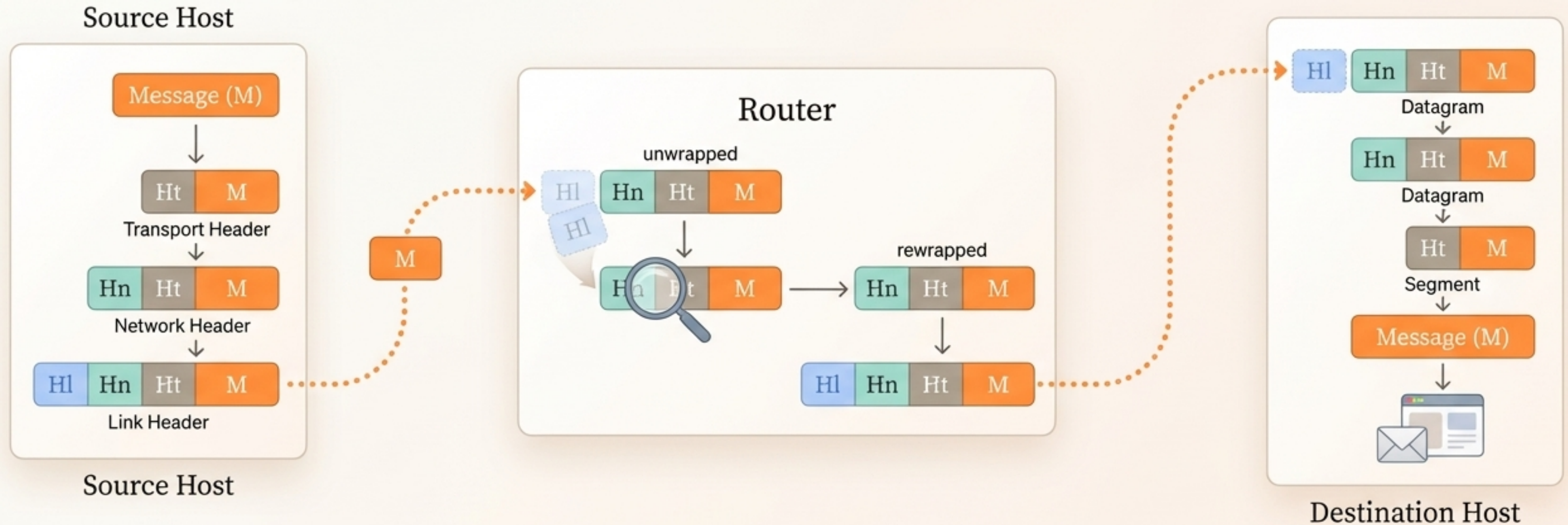


The 5-layer Internet Protocol Stack:

- **Application:** Supports network applications (HTTP, SMTP). Creates the original message.
- **Transport:** Process-to-process data transfer (TCP, UDP). Adds 'Ht'.
- **Network:** Source-to-destination routing of datagrams (IP). Adds 'Hn'.
- **Link:** Data transfer between neighboring network elements (Ethernet, WiFi). Adds 'Hl'.
- **Physical:** Bits on the wire.

Each layer adds its own header information—*encapsulating* the data from the layer above—to perform its specific function.

From a Click to Content: The Packet's Journey, Complete.



Our packet's journey—from message to frame, across access networks and the core, navigating routers, enduring delays, and guided by protocols—is the hidden story behind every interaction on the Internet. A symphony of complex systems, all working to deliver a single piece of data.